Investigating the Effect of Feedback Modality on Stringed Instrument Tuning Speed and Precision

1 Introduction

Players of stringed instruments such as the violin, cello, and guitar tune their instruments using either Type A **visual feedback** (using a digital tuner that displays deviation from the target pitch) or Type B **auditory feedback** (listening to a reference pitch and adjusting the tune by ear)[1].

According to the Cognitive Load Theory, these two modalities may impose different processing demands: visual feedback may compete with visual attention, while auditory feedback may require finer perceptual discrimination and greater working-memory use [2]. However, despite the widespread use of both approaches in string instrument tuning, it is unclear whether one feedback modality leads to faster tuning or greater precision.

Therefore, this work proposes to investigate the effect of feedback modality on string instrument tuning time and accuracy using statistical analysis. Furthermore, since many professional musicians tune without visual aids, we aim to examine whether musical experience influences the relative effectiveness of the two tuning modalities. We hypothesize that visual feedback results in faster tuning time, while auditory feedback results in greater tuning precision only among experienced musicians.

2 Methods

2.1 Participants

Participants of this experiment will be subdivided into three categories based on the number of years they have been taking lessons regarding a string instrument—low (0–2 years of lessons), moderate (2–4 years), and high (4+ years). A minimum of 15 participants will be recruited in each category. Participants will be recruited via convenience sampling (e.g., sampling at the Engineering Science common room at the University of Toronto) by completing a short questionnaire on musical experience. For this experiment, the ukulele will be used, which is portable and easy to tune across experience levels.

2.2 Experimental Variables and Design

The complete set of variables involved in this experiment is outlined in Table 1. Since the note A4 (at a frequency of 440 Hz) is an open string, it is chosen to be the reference pitch. Since an interval of 6.4 cents is considered distinguishable, we consider the instrument to be in-tune when the A4 note played has a frequency of 440 Hz ± 6.4 cents, where the cent C is defined as:

$$C = 1200 \log_2 \left(\frac{f_2}{440 \text{ Hz}} \right) \tag{1}$$

Variable type	Name	Operational definition
Independent (categorical)	Feedback type	Two levels: (1) Type A visual feedback, (2) Type B auditory feedback.
Covariate (continuous)	Music lesson experience (years)	Self-reported years of formal music lessons, for control of baseline tuning ability.
Dependent (continuous)	Tuning time, T (s)	Time required to reach the tuning criterion (speed) of 440 Hz $\pm$6.4 cents.
Dependent (continuous)	Tuning accuracy, A (Hz)	Absolute deviation of the final tuned frequency f_2 from 440 Hz (accuracy).

Table 1: Independent and dependent variables for the proposed experiment.

for the note frequency f_2 and a reference note at 440 Hz [3, 4]. The experimental procedure for an individual participant is outlined in Figure 1.

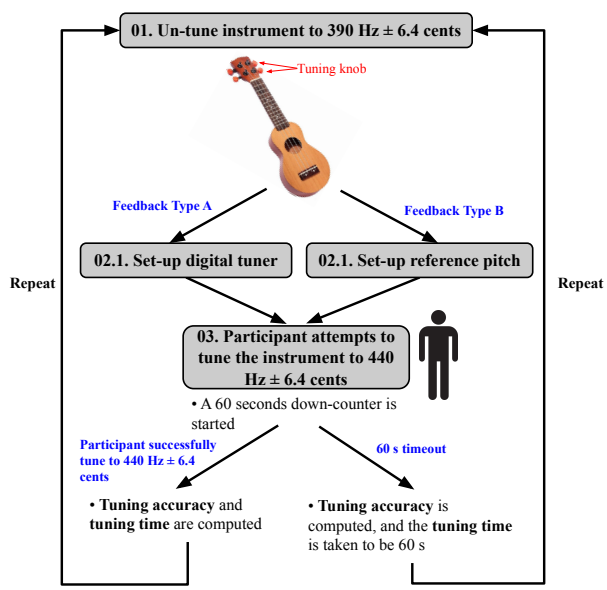

Figure 1: Flowchart of experimental procedure for each participant.

2.3 Limitations

Several limitations pertain to the current method. First, musical experience is better captured by hours of practice rather than the number of years of lessons, but accurately estimating practice time is challenging. Second, individual talents may inevitably impose an effect outside of lessons or practice.

3 Analysis Plan

3.1 Descriptive Statistics and Plots

For each feedback type, the mean, median, and standard deviation for both dependent variables, both across all participants as well as for each individual category, will be computed. Outcomes will be visualized using box-and-whisker plots and/or bar plots with error bars.

3.2 Inferential Tests

We will compare auditory vs. visual conditions on each dependent variable using an appropriate test (e.g., a two-sample t-test if assumptions are met; otherwise, a nonparametric alternative). If the experience group is included as a factor, an ANOVA or regression model can be used to control for experience.

References

[1] "Effects of visual and auditory feedback in violin and singing voice pitch matching tasks," *Frontiers in Psychology*, 2021.

[2] The Education Hub, "An introduction to cognitive load theory," https://theeducationhub.org.nz/an-introduction-to-cognitive-load-theory/, n.d., accessed: 2026-02-15.

[3] H-Pi Instruments, "Hunt System Scale," https://musictheory.zentral.zone/huntsystem2.html, 2026, zentral.zone, Accessed: Feb. 15, 2026.

[4] HyperPhysics, "The use of cents for expressing musical intervals," http://hyperphysics.phy-astr.gsu.edu/hbase/Music/cents.html, n.d., accessed: Feb. 15, 2026.